4 **(a)** longitudinal: vibrations/oscillations (of the particles/wave) are parallel to the direction **or** in the same direction (of the propagation of energy) B1

transverse: vibrations/oscillations (of the particles/wave) are perpendicular to the direction (of the propagation of energy) B1 [2]

(b) LHS: intensity = power / area units: $kg\,m\,s^{-2} \times m \times s^{-1} \times m^{-2}$ or $kg\,m^2 s^{-3} \times m^{-2}$ B1

RHS: units: $m\,s^{-1} \times kg\,m^{-3} \times s^{-2} \times m^2$ M1

LHS and RHS both $kg\,s^{-3}$ A1 [3]

(c) **(i)** change/difference in the <u>observed</u>/<u>apparent</u> frequency when the source is moving (relative to the observer) B1 [1]

(ii) wavelength increases/frequency decreases/red shift B1 [1]

(d) observed frequency = $vf_S / (v - v_S)$ C1

$550 = (340 \times 510) / (340 - v_S)$ C1

$v_S = 25\ (24.7)\ m\,s^{-1}$ A1 [3]